

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

Крючков Дмитрий Алексеевич

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
dakruchkov@dakruchkov:~$ ls /etc/ > file.txt
dakruchkov@dakruchkov:~$ ls >> file.txt
dakruchkov@dakruchkov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

Рисунок 1: Запись в файл

```
dakruchkov@dakruchkov:~$ grep .conf file.txt > conf.txt
dakruchkov@dakruchkov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage03/report/_resources/csl
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage03/report/bib/cite.bib
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage04/report/_resources/csl
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage04/report/bib/cite.bib
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage05/report/_resources/csl
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage05/report/bib/cite.bib
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage06/report/_resources/csl
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/s
tage06/report/bib/cite.bib
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/repor
t/_resources/csl
/home/dakruchkov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/repor
t/bib/cite.bib
/home/dakruchkov/git-extended/.git/hooks/commit-msg.sample
/home/dakruchkov/git-extended/.git/objects/ed/c39e317b2ea41a9df27baafc0ea30455e3cf82
/home/dakruchkov/git-extended/.git/objects/74/c5f47c664ebf6cf6ae064126bbfb8054366090
/home/dakruchkov/git-extended/.git/config
/home/dakruchkov/conf.txt
dakruchkov@dakruchkov:~$
```

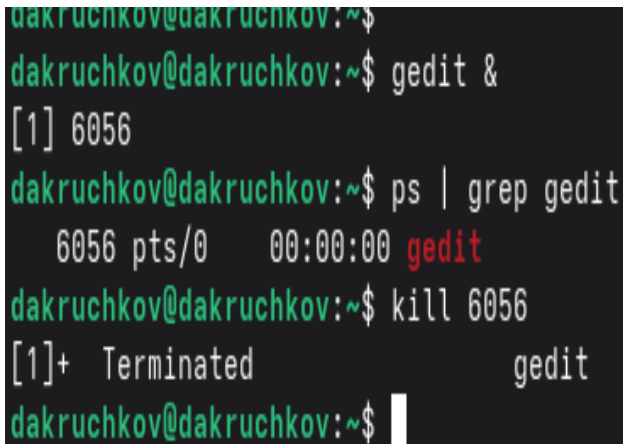
Рисунок 3: Поиск файлов

```
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss.d': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
/etc/systemd/system/multi-user.target.wants/happd.service
/etc/systemd/system/happd.service
/etc/udev/hwdb.d
/etc/udev/hwdb.bin
/etc/host.conf
:
```

Рисунок 4: Поиск файлов

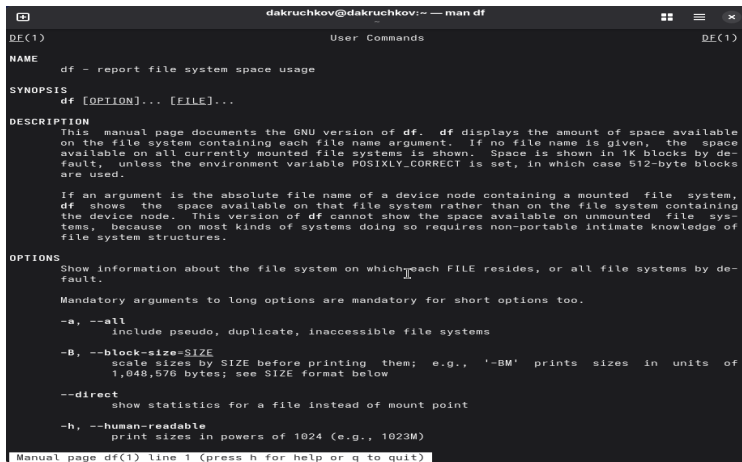
```
dakruchkov@dakruchkov:~$  
dakruchkov@dakruchkov:~$ find /etc -name "h*" -print | less  
dakruchkov@dakruchkov:~$  
dakruchkov@dakruchkov:~$ find ~ -name "log*" > logfile &  
[1] 6011  
dakruchkov@dakruchkov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
dakruchkov@dakruchkov:~$ rm logfile  
dakruchkov@dakruchkov:~$
```

Рисунок 5: Фоновый запуск процесса



```
dakruchkov@dakruchkov:~$  
dakruchkov@dakruchkov:~$ gedit &  
[1] 6056  
dakruchkov@dakruchkov:~$ ps | grep gedit  
    6056 pts/0    00:00:00 gedit  
dakruchkov@dakruchkov:~$ kill 6056  
[1]+  Terminated                  gedit  
dakruchkov@dakruchkov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса



```
dakruchkov@dakruchkov:~ — man df
DE(1)                                User Commands                                DE(1)

NAME
  df - report file system space usage

SYNOPSIS
  df [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of df.  df displays the amount of space available
  on the file system containing each file name argument.  If no file name is given, the space
  available on all currently mounted file systems is shown.  Space is shown in 1K blocks by de-
  fault, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks
  are used.

  If an argument is the absolute file name of a device node containing a mounted file system,
  df shows the space available on that file system rather than on the file system containing
  the device node.  This version of df cannot show the space available on unmounted file sys-
  tems, because on most kinds of systems doing so requires non-portable intimate knowledge of
  file system structures.

OPTIONS
  Show information about the file system on which each FILE resides, or all file systems by de-
  fault.

  Mandatory arguments to long options are mandatory for short options too.

  -a, --all
      include pseudo, duplicate, inaccessible file systems

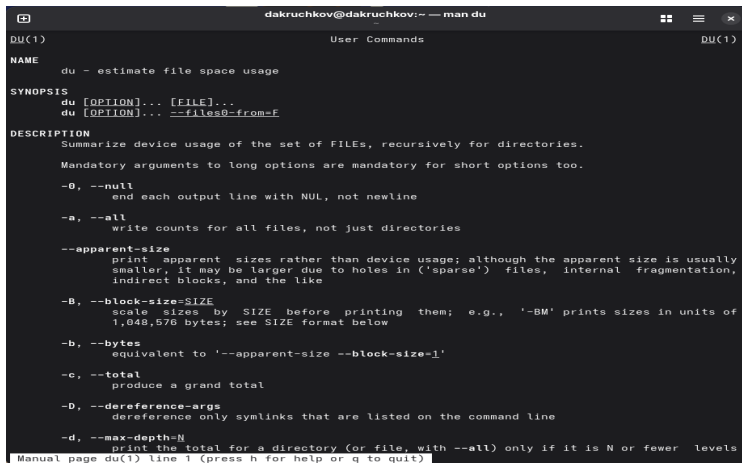
  -B, --block-size=SIZE
      scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of
      1,048,576 bytes; see SIZE format below

  --direct
      show statistics for a file instead of mount point

  -h, --human-readable
      print sizes in powers of 1024 (e.g., 1023M)

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df



```
dakruchkov@dakruchkov:~ — man du
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually
        smaller, it may be larger due to holes in ('sparse') files, internal fragmentation,
        indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of
        1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

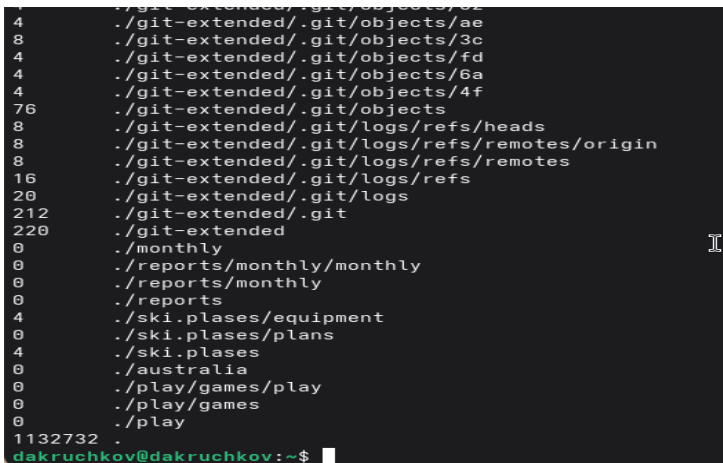
    -D, --dereference-args
        dereference only symlinks that are listed on the command line

    -d, --max-depth=N
        print the total for a directory (or file, with --all) only if it is N or fewer levels
        Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
dakruchkov@dakruchkov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 42441276 105997636 29% /
devtmpfs         4011296      0    4011296  0% /dev
tmpfs            4035668      92    4035576  1% /dev/shm
tmpfs            1614268     1948    1612320  1% /run
tmpfs             1024        0        1024  0% /run/credentials/systemd-journald.service
tmpfs            4035672    123576    3912096  4% /tmp
/dev/nvme0n1p3 155186176 42441276 105997636 29% /home
/dev/nvme0n1p2 1992552    518800    1352512 28% /boot
/dev/loop0        68480     68480        0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1       115712    115712        0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop2        49280     49280        0 100% /var/lib/snapd/snap/snapd/25935
tmpfs             1024        0        1024  0% /run/credentials/systemd-resolved.service
tmpfs            807132      148    806984  1% /run/user/1095
dakruchkov@dakruchkov:~$
```

Рисунок 9: Справка по команде du



```
1 ./git-extended/.git/objects/02
4 ./git-extended/.git/objects/ae
8 ./git-extended/.git/objects/3c
4 ./git-extended/.git/objects/fd
4 ./git-extended/.git/objects/6a
4 ./git-extended/.git/objects/4f
76 ./git-extended/.git/objects
8 ./git-extended/.git/logs/refs/heads
8 ./git-extended/.git/logs/refs/remotes/origin
8 ./git-extended/.git/logs/refs/remotes
16 ./git-extended/.git/logs/refs
20 ./git-extended/.git/logs
212 ./git-extended/.git
220 ./git-extended
0 ./monthly
0 ./reports/monthly/monthly
0 ./reports/monthly
0 ./reports
4 ./ski.plases/equipment
0 ./ski.plases/plans
4 ./ski.plases
0 ./australia
0 ./play/games/play
0 ./play/games
0 ./play
1132732 .
dakruchkov@dakruchkov:~$
```

Рисунок 10: Запуск команды du


```
/home/dakruchkov/git-extended/.git/objects/7e
/home/dakruchkov/git-extended/.git/objects/f0
/home/dakruchkov/git-extended/.git/objects/62
/home/dakruchkov/git-extended/.git/objects/ae
/home/dakruchkov/git-extended/.git/objects/3c
/home/dakruchkov/git-extended/.git/objects/fd
/home/dakruchkov/git-extended/.git/objects/6a
/home/dakruchkov/git-extended/.git/objects/4f
/home/dakruchkov/git-extended/.git/logs
/home/dakruchkov/git-extended/.git/logs/refs
/home/dakruchkov/git-extended/.git/logs/refs/heads
/home/dakruchkov/git-extended/.git/logs/refs/remotes
/home/dakruchkov/git-extended/.git/logs/refs/remotes/origin
/home/dakruchkov/monthly
/home/dakruchkov/reports
/home/dakruchkov/reports/monthly
/home/dakruchkov/reports/monthly/monthly
/home/dakruchkov/ski.places
/home/dakruchkov/ski.places/equipment
/home/dakruchkov/ski.places/plans
/home/dakruchkov/australia
/home/dakruchkov/play
/home/dakruchkov/play/games
/home/dakruchkov/play/games/play
dakruchkov@dakruchkov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.